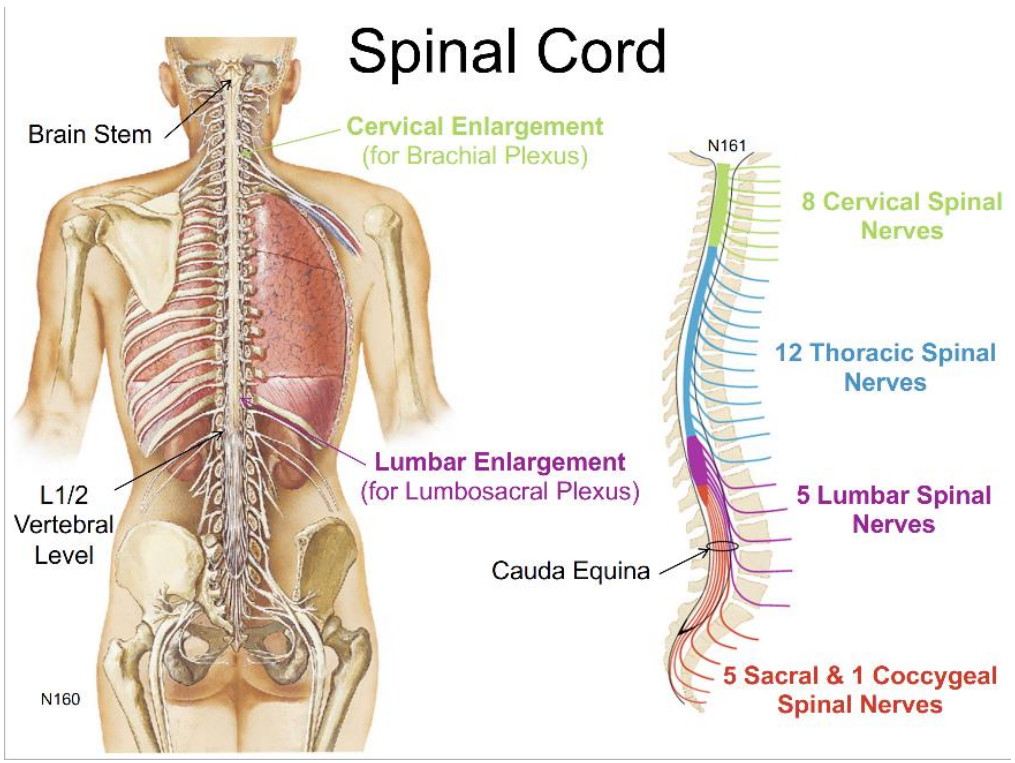
Week #8: [Non-Mechanical Back & Neck Pain]

Clinical Approach to Non-Mechanical Back Pain

Non-Mechanical Axial Pain **(Dr. Caitlin Cassidy)**

Clinical Approach to Non-Mechanical Back Pain

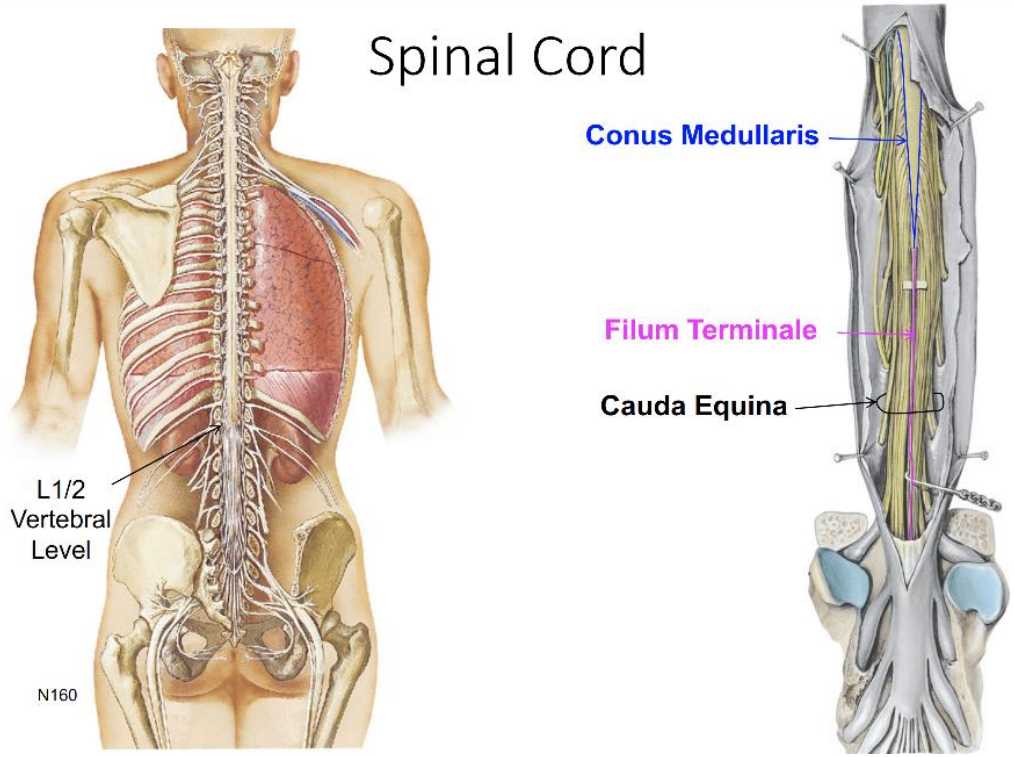


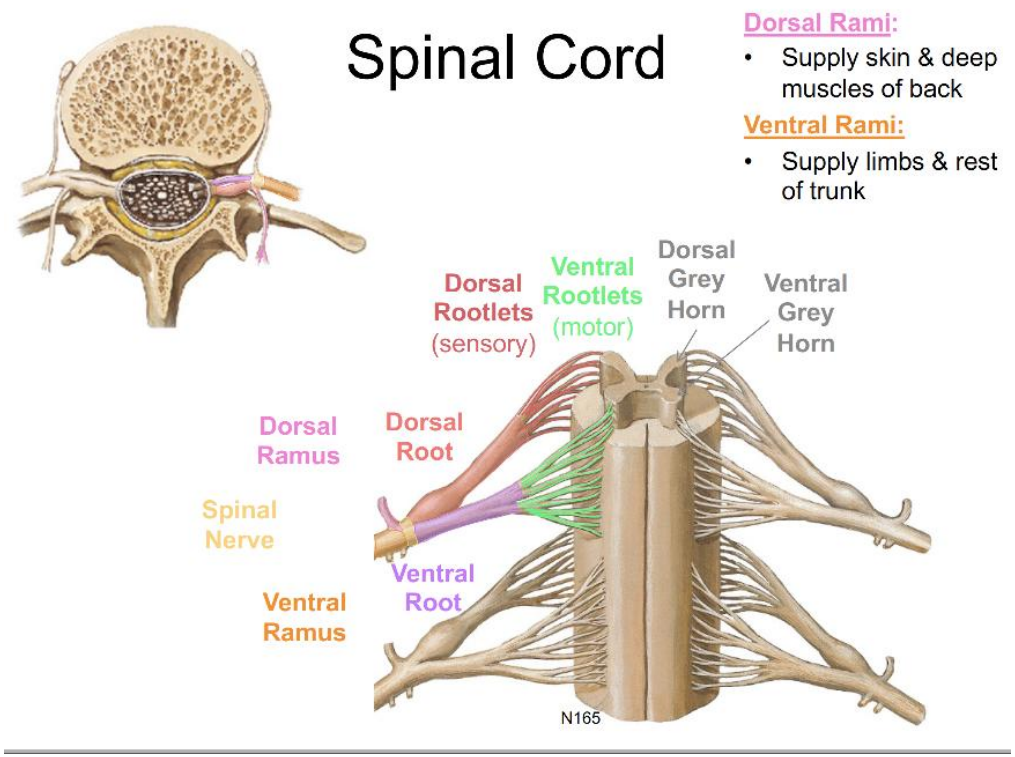
Anatomy of the Spinal Cord:

a. Part of the **central nervous system (CNS)** and continuous with the brainstem (medulla). Major reflex center and conduction pathway between the body and the brain
b. Housed within the vertebral column, it extends from the foramen magnum to the **L1-L2 vertebral level** in adults
c. Diameter not uniform
    i. **Cervical Enlargement** – C4-T1 spinal cord levels; associated with the brachial plexus, which innervates the entire upper limb
    ii. **Lumbosacral Enlargement** – L1-S3 spinal cord levels; associated with lumbar and sacral plexuses, which innervates the entire lower limb
    iii. **Conus Medullaris** – tapering inferior end of the spinal cord that terminates inferior to the lumbar enlargement
    iv. **Filum Terminale Internum** – continuation of the pia mater; extends as slender, anchoring filament from the tip of the conus medullaris to the coccyx

Spinal Nerve Organization:

a. 31 pairs of spinal nerves:
    i. 8 cervical
    ii. 12 thoracic
    iii. 5 lumbar
    iv. 5 sacral
    v. 1 coccygeal





In adults, the spinal cord is shorter than the vertebral column. Originally in the embryo, the spinal cord extended the full length of the spinal canal, so spinal cord segments were located at the level of the corresponding vertebra. However, growth of the spinal cord does not keep pace with the growth of the vertebral column, such that the spinal cord eventually only occupies the superior 2/3 of the spinal canal.

1. **Cervical segments** remain close to their corresponding vertebrae. **C1-C7 spinal nerves** exit superior to their corresponding vertebrae, while the C8 spinal nerve exits below the C7 vertebrae
2. Below the cervical region, spinal nerves exit the vertebral column by passing inferior to the pedicles of the corresponding vertebrae
3. The distance between a spinal cord segment giving rise to a pair of spinal nerves and the intervertebral foramen through which the nerves must exit the vertebral column increases as the cord descends
4. While the spinal cord ends at ~L1/2 in adults, nerve roots that branch off the lower lumbar and sacral regions of the spinal cord must still descend to exit the ***vertebral column at their respective intervertebral foramina***
5. The nerve roots continuing past the termination of the spinal cord comprise the cauda equina, which means “horse’s tail” due to the appearance of all the nerve roots extending inferiorly

Multiple **rootlets** emerge from ventral (anterior) and dorsal (ventral) sides of the spinal cord that merge to form:

1. **Ventral roots** of spinal nerves
    a. Contain **motor** (efferent) **fibres** and **presynaptic autonomic fibres**
    b. Cell bodies of axons contributing to the ventral roots are located within the **anterior horn of the gray matter** within the spinal cord
2. **Dorsal roots** of spinal nerves
    a. Contain **sensory** (afferent) **fibres**
    b. Cell bodies of axons contributing to dorsal roots are located in the **spinal ganglia** (dorsal root ganglia) that appears as a small swelling on dorsal root

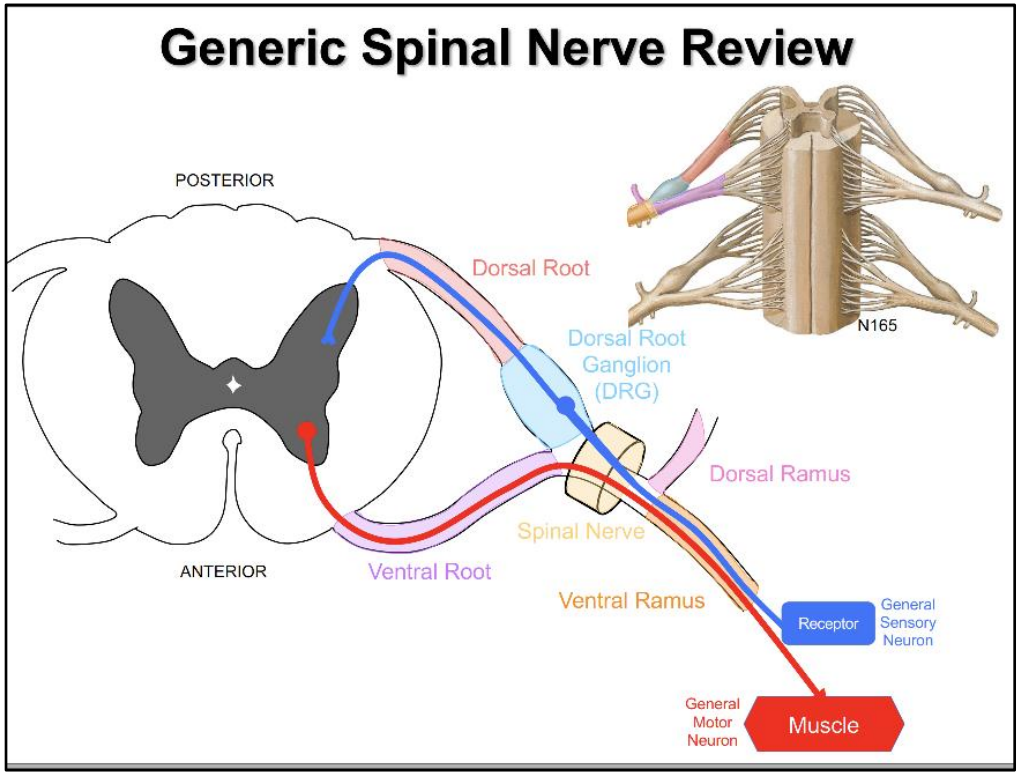
Ventral and dorsal roots unite to form a mixed (carries sensory and motor innervation) **spinal nerve** within the **intervertebral foramen**

Each spinal nerve divides into two rami:

1. **Dorsal Ramus** – supplies the skin and deep muscles of the back
2. **Ventral Ramus** – supplies the limbs and the rest of the trunk

Clinical Note: The lumbar spinal nerves increase in size from superior to inferior, while the lumbar intervertebral foramina decrease in diameter. Thus, the more inferior nerves are more vulnerable to compression (L5 spinal nerves are the largest, while their intervertebral foramina are the smallest).

Each pair of spinal nerves represents a specific segment of the spinal cord and provides sensory innervation for a specific area of skin (**dermatome**) and motor innervation to a specific group of muscles (**myotome**)

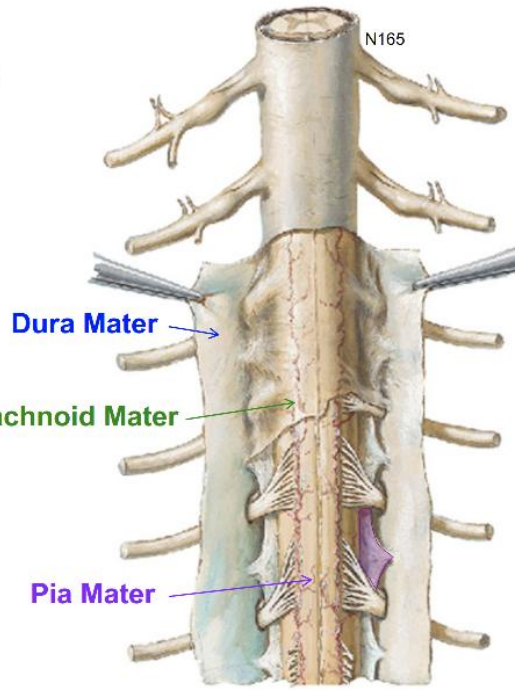


Somatic Pathway:

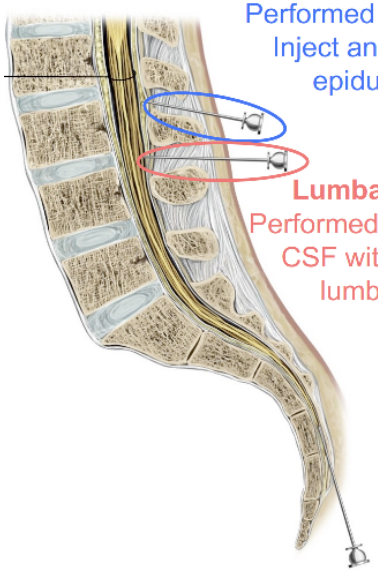
a. **General Sensory (Afferent) Neuron** - Transmits sensations from a receptor on the body wall through the spinal nerve and dorsal root to the CNS. Cell bodies for these neurons are located within the dorsal root ganglia (DRG).
b. **Synapse** – Once inside the spinal cord, the sensory fibres enter the dorsal grey horns where they will synapse with a secondary neuron, which will transmit the impulse to the brain.
c. **General Motor (Efferent) Neuron** – leaves the spinal cord through the **ventral root**, enters the **spinal nerve** and passes along motor branches of either the **anterior** or **posterior primary rami** to reach its target (skeletal muscles). Cell bodies for these neurons are located within the ventral grey horn of the spinal cord.
    i. **Reflex Arc** – sensory fibres may either pass directly to the ventral grey horn or they synapse with an interneuron, which passes to the ventral grey horn. In this situation, the sensory neuron synapses directly with the motor neuron, bypassing the brain.

Meninges

- Continuous with the meninges of the brain
- Surround, support, & protect
- **Dura Mater** "tough mother"
  - Forms dural sac (to S2)
- **Arachnoid Mater** "web like"
  - Delicate, avascular layer
- **Pia Mater:**
  - Innermost layer
  - **Denticulate ligaments** anchor spinal cord
  - Continues as filum terminale



Meninges



- **Epidural Space**
  - Real space
  - Superficial to dura
  - Occupied by fat & venous plexus
- **Subdural Space**
  - Potential space
  - Between dura & arachnoid
- **Subarachnoid Space**
  - Real space
  - Contains **cerebrospinal fluid (CSF)** & vasculature

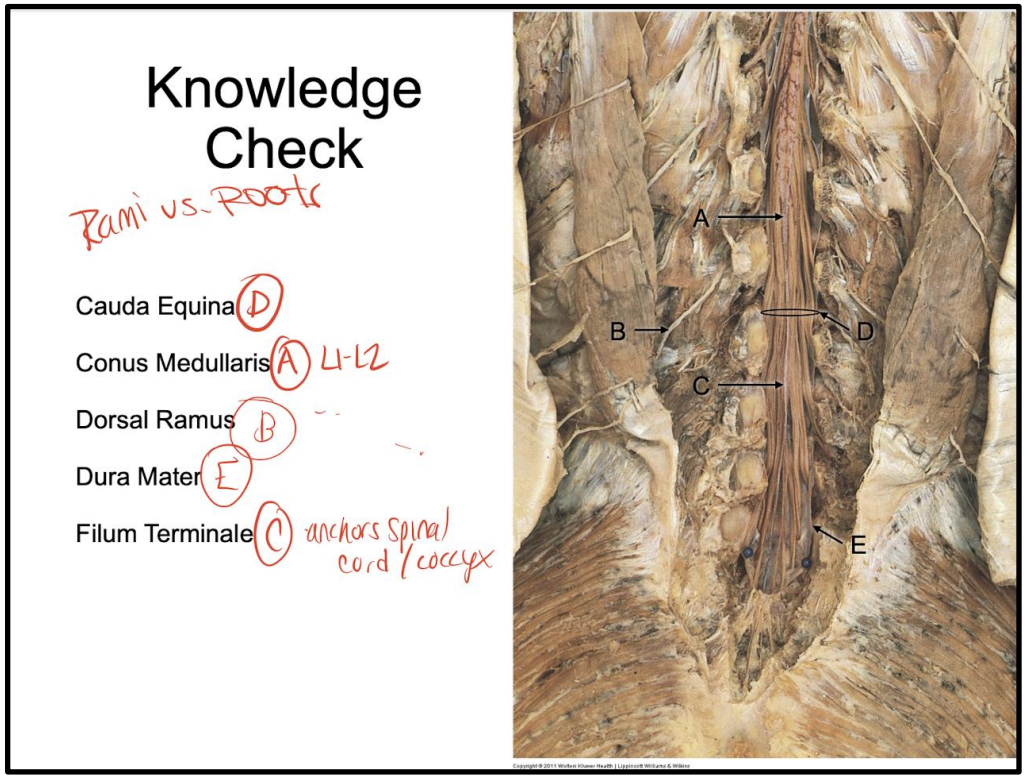
Spinal Meninges:

a. Membranes that surround, support, and protect the spinal cord and spinal nerve roots
b. Three layers:
   i. **Spinal Dura Mater ("tough mother")**
      1. Tough outermost covering of spinal cord
      2. Forms **spinal dural sac**, a long tubular sheath within vertebral canal
         a. Extends inferiorly to the S2 vertebral level
         b. Pierced by spinal nerves
         c. Anchored to the coccyx by the **filum terminale externum**
         d. Continuous with cranial dura mater at foramen magnum
      3. **Epidural Space** – actual space superficial to dura mater; occupied by fat and the internal vertebral venous plexus
      4. **Subdural Space** – potential space between the dura and arachnoid mater
      5. **Clinical Note:** During an **epidural block,** anesthetic can be injected into the epidural space, which would directly affect the spinal nerve roots of the cauda equina after they exit from the dural sac. The patient would lose sensation inferior to the level of the block. This is frequently used during labor and delivery.
   ii. **Spinal Arachnoid Mater**
      1. Delicate, avascular membrane (middle layer)
      2. Encloses **subarachnoid space** containing **cerebrospinal fluid** (CSF) and vasculature to the spinal cord
      3. Not attached to dura mater, but held against it by pressure from the CSF

4. **Clinical Note:** A **lumbar spinal puncture** is performed to obtain a CSF sample from the **lumbar cistern**, an enlargement of the subarachnoid space between the termination of the spinal cord (L2 level) and termination of the dural sac (S2 level) that houses the cauda equina. A needle is inserted into the subarachnoid space while the patient is lying on his/her side with the trunk flexed. It is usually performed at the L3/L4 or L4/L5 vertebral level.

iii. Spinal Pia Mater

1. Innermost covering membrane directly covering the spinal cord, nerve roots, and vessels
2. **Denticulate ligaments** – tooth-like extensions extend between anterior and posterior roots that anchor the spinal cord in the dural sac
3. Continues as **filum terminale internum** inferior to the conus medullaris



Additional Notes from Text Notes:

C. Muscular System:

a. The **myotome** portion of the **dermomyotome** gives rise to the muscles of the axial skeleton, body wall, and limbs

b. By the end of the fifth week, each typical myotome divides into a dorsal **epimere (epaxial division)** and a ventral **hypomere (hypaxial division)**

i. **Myoblasts** from the **epimere** form the **epaxial muscles.**

1. Splenius muscles, erecter spinae muscles, and the transversospinalis muscle group

ii. **Myoblasts** from the **hypomere** form the **hypaxial muscles.**

1. Lateral and ventral muscle of the trunk and all of the muscles of the limbs

c. Derivatives of each **somite** are supplied by the nerve that originally served that s**omite**. Thus, as muscle differentiate into epimeres and hypomeres, each spinal nerve divides into a **dorsal primary ramus**, to serve **epimere-derived** muscles from that segment, and a **ventral primary ramus** to innervate **hypomere-derived** muscles from that segment

i. Some muscles, such as the intercostal muscles, remain segmentally arranged but most myotomes fusewith others to form individual skeletal muscles;

thus, most muscles are innervated by more than one spinal segment

1. This occurs by innervation to a muscle from multiple spinal nerves (i.e. muscles of the trunk) or the joining of multiple spinal nerves into single nerves in the brachial and lumbosacral plexuses for limb muscle innervation

D. Cutaneous Innervation:

1. Recall, each somite is served by a single spinal nerve
2. During the fifth week, **sensory peripheral nerves** grow from their developing limb plexuses (brachial and lumbosacral), using the motor nerves as their 'guides' into the mesenchyme of the limbs
3. The spinal nerves are distributed in segmental bands called **dermatomes.** A **dermatome** is the area of skin supplied by a single spinal nerve and its dorsal ganglion
    a. However, dermatomes considerably overlap one another
4. Even when muscle cells migrate, within the elongating limb, they continue to be innervated by the spinal nerve from their segment of origin
    a. This explains the **segmental pattern** of innervation of the muscles (myotomes) and the skin **(dermatomes)**
    b. Note there is minimal dermatome overlap where non-sequential spinal segments are adjacent at the **ventral axial line**

Non-Mechanical Axial Pain

- Neurological edition: Problems to the side -radiculopathy
    - Problems to the side - radiculopathy
    - Problems to the centre
        - LUMBAR (RED FLAGS)
        - CERVICAL (RED FLAGS)
    - Everything Else - RED FLAGS
        - Inflammatory
        - Fractures
        - Visceral
        - Infection
        - Neoplasm
    - **Done by Dr. Caitlin Cassidy**